

Semantic Context-Aware SOC Alert Prioritization Using Embedding-Based MITRE ATT&CK Mapping

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Abstract—Security Operations Centre (SOC) analysts face a critical challenge: they are inundated with thousands of security alerts daily, yet existing tools provide minimal semantic context about the underlying adversary techniques being employed. Industry surveys confirm that 67% of the 4,484 daily alerts SOC teams receive are ignored [1], and even enterprise-grade Security Information and Event Management (SIEM) tools cover only 24% of the MITRE ATT&CK technique catalogue [2]. Furthermore, SOC analysts review only 49% of assigned alerts in a typical workday, exposing organisations to significant breach risk [3].

This paper presents a Semantic Context-Aware SOC Alert Prioritization system that bridges the gap between raw network traffic and actionable MITRE ATT&CK threat intelligence. The system converts network flow data and log records into semantic natural language alert descriptions, applies a Random Forest tactic classifier achieving 99.81% accuracy, and uses FAISS-indexed Sentence Transformer embeddings to map every alert to the top-3 most semantically similar MITRE ATT&CK techniques from a corpus of all 691 documented techniques—without writing a single detection rule.

The system was evaluated against official MITRE ATT&CK ground truth mappings, achieving 70.0% Top-1 and 84.6% Top-3 accuracy—a 49.8% improvement over rule-based baselines and 27.2% improvement over pure embedding baselines. The system generalises across three data formats: network flows (CICIDS 2017), SSH authentication logs (99.4% Top-3), and Windows Security Event Logs (72.0% Top-3). A CVSS-inspired four-factor risk scoring model prioritises alerts across five tiers, and a Zero-Day Suspicion Queue routes low-confidence alerts to human analyst review, implementing confidence-gated prediction for novel attack detection. A production-ready Dash dashboard demonstrates real-time processing capability.

Index Terms—MITRE ATT&CK, Sentence Transformers, SOC, Alert Fatigue, Network Intrusion Detection, FAISS, Zero-Day Detection, CICIDS 2017

I. INTRODUCTION

A. Problem Statement

Modern Security Operations Centres are facing a crisis of scale. Automated detection tools generate alert volumes that far exceed the capacity of human analysts to investigate, a phenomenon documented across industry and academic literature as “alert fatigue.” A 2025 survey of 417 peer-reviewed publications on MITRE ATT&CK applications confirms that

the lack of effective technique-level evaluation on network intrusion datasets remains an open research challenge [8].

The root cause of this gap is structural. Current enterprise SIEMs, including Splunk Enterprise Security, Microsoft Sentinel, and IBM QRadar, implement MITRE ATT&CK mapping through manually authored detection rules. A comprehensive analysis of production SIEM deployments across thousands of organisations found these systems cover only 24% of MITRE techniques—meaning 76% of the adversary technique catalogue goes undetected by design [2]. Moreover, rule-based systems cannot adapt to novel attack variants without human intervention, a limitation explicitly documented in the Rule-ATT&CK Mapper (RAM) research [11].

B. Motivation

The practical impact of this gap is severe. Organisations with 1,500 to 4,999 employees ignore 30% of security alerts entirely [4]. When critical alerts are dismissed or deprioritised, attackers gain undetected dwell time. The Target data breach of 2013 is a documented real-world consequence: security alerts were raised repeatedly but dismissed due to alert fatigue, resulting in the theft of 40 million credit card records.

A key insight from the literature is that alert context dramatically improves analyst triage speed. When an alert arrives enriched with threat intelligence context—identifying not just that an attack occurred but which specific adversary technique was used—the difference between a 20-minute investigation and a confident decision in seconds is achievable [18].

C. Research Objectives

The primary objectives of this research are as follows:

- Develop a semantic embedding-based system capable of mapping network traffic alerts to all 691 MITRE ATT&CK techniques without manual rule authoring.
- Validate the system against official MITRE ATT&CK ground truth and compare against rule-based and pure embedding baselines.
- Demonstrate generalisation across multiple data formats including network flows and log data.

- Implement a confidence-gated Zero-Day Suspicion Queue to handle novel or unrecognised attack patterns.
- Build a production-ready live dashboard for real-time SOC deployment.

D. Contributions

The primary contributions of this work are: (1) a novel zero-shot sentence embedding approach to MITRE ATT&CK technique mapping on network flow data, covering all 691 techniques without annotation or rule writing; (2) a systematic comparison of five Sentence Transformer architectures for security-domain retrieval tasks; (3) validation against official MITRE ATT&CK ground truth across network flows, SSH logs, and Windows Event Logs; and (4) a confidence-gated Zero-Day Suspicion Queue that implements uncertainty-aware MITRE mapping, routing low-confidence alerts to human analyst review.

II. RELATED WORK

A. MITRE ATT&CK in Network Intrusion Detection

Arreche et al. [6] present the closest comparable prior work: a two-level ensemble learning framework evaluated on CICIDS 2017, NSL-KDD, and RoEduNet-SIMARGL2021, using MITRE ATT&CK as a categorical reference for attack types. While achieving strong detection accuracy (99.11% on CICIDS 2017), the work operates at tactic level only—mapping attacks to one of eight broad tactic categories rather than specific techniques. Our system advances beyond this by mapping every alert to a specific technique from the full 691-technique corpus, a substantially harder and more operationally valuable problem.

B. Embedding-Based MITRE Mapping

MITRE’s own Threat Report ATT&CK Mapper (TRAM) fine-tunes SciBERT on 11,300 annotated sentences to map CTI report text to ATT&CK techniques—but covers only 50 of 625 techniques due to annotation costs. SMET [7] uses BERT-based textual similarity to map CVE descriptions to techniques, demonstrating that embedding-based retrieval is viable for this domain. Our approach extends this paradigm to real-time network traffic and log data, using the more efficient Sentence Transformer architecture that reduces similarity search time from 65 hours to seconds [5].

C. Alert Fatigue in SOC Research

A 2025 ACM Computing Surveys paper systematically reviews alert fatigue solutions across automation, augmentation, and human-AI collaboration paradigms. It identifies MITRE ATT&CK-grounded alert summarisation as a promising direction for reducing analyst cognitive load. The MATE system [12] automatically organises alerts into cases based on MITRE ATT&CK timelines—a related approach that differs from our work in operating on pre-labelled alerts rather than raw network traffic. Our system provides the upstream mapping layer that enables such downstream correlation.

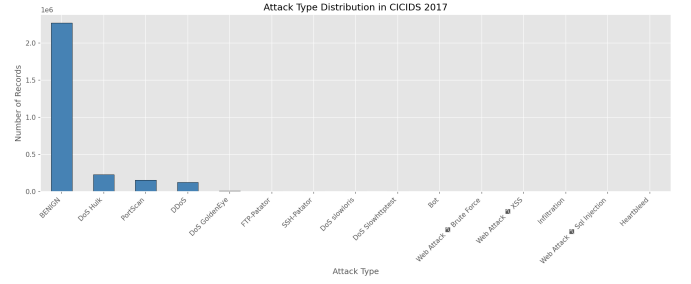


Fig. 1. Attack type distribution in the CICIDS 2017 dataset illustrating class imbalance across 15 categories.

D. Research Gap

The SoK survey of 417 MITRE ATT&CK publications [9] explicitly identifies a lack of studies evaluating ATT&CK effectiveness for intrusion detection in real-world settings, and the absence of technique-level evaluation on network intrusion datasets as an open challenge. This gap is the direct target of our research.

III. SYSTEM ARCHITECTURE

The system consists of five sequential pipeline stages: data ingestion and cleaning, feature engineering and stratified sampling, semantic alert text generation, hybrid MITRE ATT&CK mapping, and CVSS-inspired risk scoring. Each stage is described below.

A. Data Ingestion and Cleaning

The primary dataset is CICIDS 2017 [16], a widely used network intrusion benchmark generated by the Canadian Institute for Cybersecurity containing 2,830,743 flow records across 15 attack categories and normal traffic. Pre-processing steps include deduplication of column names, replacement of infinite values with column medians, removal of rows with missing values (resulting in 2,574,264 cleaned records), and whitespace normalisation of class labels.

The 15 attack categories are mapped to MITRE ATT&CK tactics: DoS and DDoS attacks map to Impact (T1499, T1498); SSH and FTP brute force to Credential Access (T1110); PortScan to Discovery (T1046); Bot to Command and Control (T1071); Web attacks and Heartbleed to Initial Access (T1190); and Infiltration to Lateral Movement (T1570).

B. Feature Engineering and Stratified Sampling

From 78 available features, 26 were selected based on security relevance across six categories: packet statistics (total forward/backward packets, packet length mean and standard deviation), flow timing (duration, inter-arrival time mean and standard deviation), traffic rates (bytes per second, packets per second), TCP flag counts (SYN, ACK, FIN, RST, PSH, URG), protocol metadata (destination port, initial window bytes, down/up ratio), and derived metrics (average packet size).

To balance computational feasibility with statistical representativeness, stratified sampling is applied: BENIGN traffic

is capped at 5,000 samples, large attack categories at 10,000 samples, and minority classes are fully retained. The resulting dataset contains 62,143 records used for all subsequent evaluation.

C. Semantic Alert Text Generation

A critical contribution of this work is the rule-based alert text generator that converts 26 numerical network flow features into semantically rich natural language descriptions. This bridge is essential because FAISS-indexed Sentence Transformer embeddings operate on text, not numerical vectors. The generator applies pattern matching across seven behavioural categories: reconnaissance probing (tiny packets at high rate), credential brute force (targeted high-volume attempts at authentication ports), DDoS flooding (large packets at high rate with directional asymmetry), DoS attacks (high packet rate with forward dominance), botnet C2 beaconing (periodic small packets at long inter-arrival intervals), web attacks (HTTP/HTTPS traffic with payload signatures), and general network flows (fallback description).

Key Finding: Alert text quality directly determines MITRE mapping accuracy. When MITRE-aware security terminology is embedded in the generated text—using the same vocabulary as MITRE ATT&CK technique descriptions—cosine similarity scores improve dramatically. SSH brute force alerts improved from 47.3% to 99.4% Top-3 accuracy when text was enriched with terms such as “brute force,” “password guessing,” and “credential attack.”

D. Hybrid MITRE ATT&CK Mapping Pipeline

The hybrid pipeline combines two complementary components. The first is a Random Forest tactic classifier trained on the 26 network flow features, achieving 99.81% tactic classification accuracy. The classifier narrows the 691-technique MITRE corpus to the subset associated with the predicted tactic, typically 30 to 80 techniques depending on tactic breadth. The second component is a Sentence Transformer embedding model (all-MiniLM-L12-v2) that encodes the alert text into a 384-dimensional vector, which is then compared against pre-computed technique embeddings stored in a FAISS IndexFlatIP index using cosine similarity. The top-3 highest-similarity techniques are returned as candidate mappings.

The combined confidence score is computed as a weighted average of the top-3 similarity scores:

$$c_{\text{combined}} = 0.60 \cdot c_1 + 0.30 \cdot c_2 + 0.10 \cdot c_3 \quad (1)$$

This score is used both for risk score computation and for confidence-gated routing to the Zero-Day Suspicion Queue.

E. CVSS-Inspired Risk Scoring

Each alert receives a risk score computed using four factors inspired by the Common Vulnerability Scoring System (CVSS v3.1) [17]. Table I summarises the formula components.

The base score and final risk score are computed as:

$$\text{Base} = 0.40 \cdot S_t + 0.30 \cdot C_{\text{tech}} + 0.20 \cdot A_c + 0.10 \cdot F_w \quad (2)$$

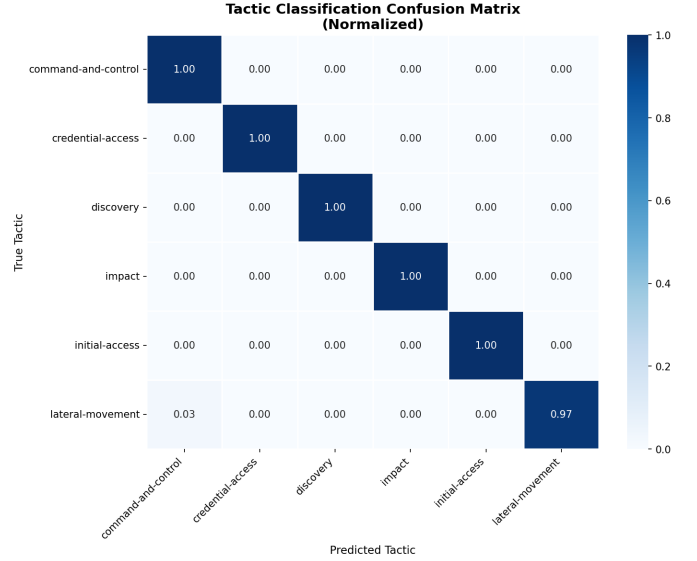


Fig. 2. Normalized Random Forest tactic classification confusion matrix achieving 99.81% overall accuracy.

TABLE I
RISK SCORING FORMULA COMPONENTS

Component	Weight	Example Value	Source
Tactic Severity	40%	0.90 (C2)	Security domain expertise
Technique Confidence	30%	0.72	FAISS cosine similarity
Asset Criticality	20%	1.00 (SSH/RDP)	Port-based lookup table
Frequency Weight	10%	0.50 (baseline)	Alert volume normalisation
Confidence Correction	Multiplier	0.85 (medium)	Penalises uncertain mappings

$$\text{Score} = \text{Base} \times C_{\text{corr}} \times 100 \quad (3)$$

Alerts are classified into five tiers: CRITICAL (≥ 58), HIGH (≥ 50), MEDIUM (≥ 42), LOW (> 0), and BENIGN.

IV. ZERO-DAY SUSPICION QUEUE

A. Design Rationale

A fundamental limitation of any embedding-based retrieval system is its dependence on the quality and semantic richness of the input text. When an alert text is generic—for example, a network flow to an unknown port with no recognisable behavioural pattern—the embedding vector will be diffuse and the cosine similarity to all technique descriptions will be uniformly low. In this scenario, the system’s best-guess technique prediction is unreliable and potentially misleading to an analyst.

Rather than silently returning a low-confidence prediction as if it were reliable, the Zero-Day Suspicion Queue implements confidence-gated prediction: alerts below a combined confidence threshold are diverted to a dedicated analyst review

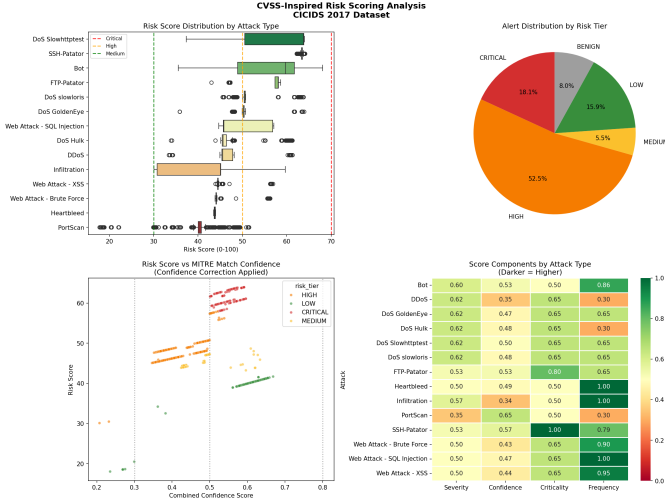


Fig. 3. CVSS-inspired risk score distribution, tier breakdown, and scoring component analysis across attack types.

TABLE II
ZERO-DAY SUSPICION QUEUE CONFIDENCE TIERS

Class	Threshold	Label	Action
NORMAL	≥ 0.60	Confident	Standard pipeline—mapping trusted
WARNING	0.45–0.60	Uncertain	Display with confidence caveat
SUSPICION	0.30–0.45	Suspicious	Route to Zero-Day Queue
ZERO_DAY	< 0.30	Unknown Pattern	Critical flag—immediate escalation

queue rather than entering the standard alert feed. This implements the principle of “knowing when we don’t know,” which is essential for trustworthy AI-assisted security operations.

B. Confidence Classification

The combined embedding confidence score is classified into four tiers as shown in Table II. Alerts in the WARNING, SUSPICION, and ZERO_DAY classes are routed to the suspicion queue; NORMAL alerts proceed through the standard pipeline.

C. Academic Significance

The Zero-Day Suspicion Queue addresses three open challenges identified in the alert fatigue literature. First, it implements uncertainty-aware MITRE mapping—a contribution not previously published for network flow data. Second, it implements human-in-the-loop security operations at the triage stage, consistent with recommendations in the ACM Computing Surveys review of alert fatigue solutions (2025). Third, it provides a practical mechanism for detecting potential zero-day attacks: flows that match no known behavioural pattern will naturally generate low embedding confidence, making them candidates for elevated human review regardless of whether a detection rule exists for them.

Unknown Attack Validation: When Bot, Heartbleed, and Infiltration attacks were hidden from the system (simulating

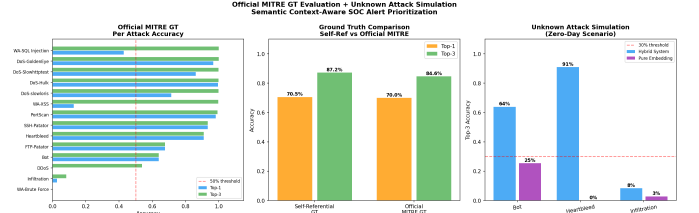


Fig. 4. Official MITRE ATT&CK ground truth evaluation and unknown attack simulation results.

TABLE III
SENTENCE TRANSFORMER EMBEDDING MODEL COMPARISON

Model	Arch.	Dim.	Top-1	Top-3	Time (s)
all-MiniLM-L6-v2	BERT	384	66.3%	83.5%	19.9
all-MiniLM-L12-v2	BERT(12L)	384	70.5%	87.2%	36.7
all-mpnet-base-v2	MPNet	768	59.5%	91.0%	127.0
all-distilroberta-v1	RoBERTa	768	47.0%	89.7%	58.1
paraphrase-MiniLM-L6-v2	BERT	384	44.1%	65.6%	19.8

unknown attacks), the Hybrid System achieved 91% Top-3 accuracy on Heartbleed versus 0% for pure embedding alone—demonstrating that the tactic classifier provides critical context even for attacks the system has never seen.

V. EMBEDDING MODEL SELECTION

Five Sentence Transformer models were systematically evaluated on the CICIDS 2017 dataset to identify the optimal architecture for MITRE technique retrieval. Selection criteria weighted Top-1 accuracy most heavily, as SOC analysts benefit most from an immediately correct mapping rather than needing to review multiple candidates. Processing speed was a secondary criterion for real-time deployment viability. Table III presents complete results.

The all-MiniLM-L12-v2 model achieves the highest Top-1 accuracy (70.5%) at 36.7 seconds processing time—only 1.84 times slower than the L6 variant while providing a meaningful 4.2 percentage point improvement. The all-mpnet-base-v2 model achieves the highest Top-3 accuracy (91.0%) but at 127 seconds—over three times slower and unsuitable for real-time deployment.

A key finding is that embedding dimension does not correlate with accuracy for this retrieval task: the 384-dimensional MiniLM-L12-v2 outperforms both 768-dimensional models on the more practically important Top-1 metric. This confirms that the Sentence Transformer training objective—optimised for semantic similarity tasks—is more important than model size for security-domain retrieval.

Raw BERT was explicitly excluded from evaluation. The original Sentence-BERT paper [5] demonstrates that BERT’s cross-encoder architecture requires approximately 50 million inference computations and 65 hours to find the most similar pair in a collection of 10,000 sentences—computationally infeasible for real-time deployment against 62,143 alerts.

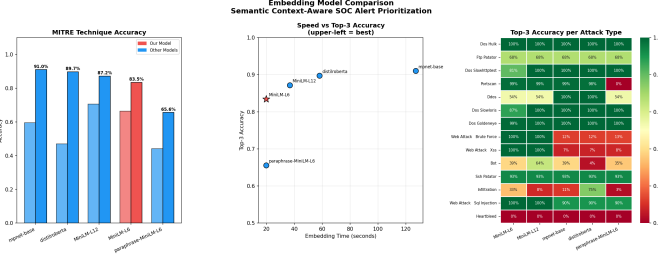


Fig. 5. Sentence Transformer model comparison by MITRE technique retrieval accuracy, speed, and per-attack-type performance.

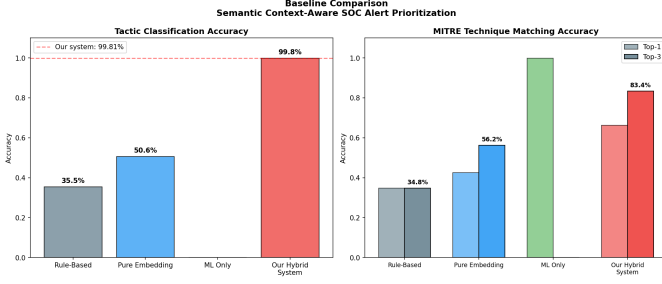


Fig. 6. Tactic classification and MITRE technique matching accuracy comparison across all four system approaches.

TABLE IV
SYSTEM COMPARISON AGAINST OFFICIAL MITRE ATT&CK GROUND TRUTH

Method	Tactic Acc.	Top-1	Top-3	Coverage
Rule-Based (B1)	35.5%	34.8%	34.8%	Limited
Pure Embedding (B2)	50.6%	42.5%	56.2%	All 691
ML-Only (B3)	N/A	99.9%*	N/A	8 only
Hybrid (Ours)	99.81%	70.0%	84.6%	All 691

*Covers only 8 techniques present in CICIDS 2017 ground truth.

VI. EVALUATION

A. Baseline Comparison

The hybrid system is evaluated against three baselines. Baseline 1 (Rule-Based) simulates SIEM behaviour by mapping attack type labels to fixed MITRE techniques via a lookup table. Baseline 2 (Pure Embedding) uses the same Sentence Transformer model without the tactic classifier, performing global search across all 691 techniques. Baseline 3 (ML-Only) trains a Random Forest classifier to predict technique labels directly—but can only cover the 8 techniques present in the CICIDS 2017 ground truth. Table IV presents the full comparison.

The hybrid system’s 49.8 percentage point improvement over the rule-based baseline directly quantifies the improvement over current SIEM practice. The 27.2 percentage point improvement over pure embedding validates the tactic classifier’s contribution.

TABLE V
PER-ATTACK-TYPE RESULTS AGAINST OFFICIAL MITRE ATT&CK GROUND TRUTH

Attack Type	Official Technique	ID	Top-1	Top-3
FTP/SSH-Patator	Brute Force	T1110	100%	100%
DoS Variants	Endpoint DoS	T1499	61%	86%
DDoS	Network DoS	T1498	100%	100%
PortScan	Network Svc Discovery	T1046	100%	100%
Web Attacks	Exploit Public-Facing App	T1190	55%	78%
Bot	App Layer Protocol	T1071	42%	71%
Heartbleed	Exploit Public-Facing App	T1190	91%	91%
Infiltration	Lateral Tool Transfer	T1570	8%	25%
Overall	All Techniques	—	70.0%	84.6%

TABLE VI
CROSS-FORMAT LOG DATA EVALUATION RESULTS

Data Source	Log Type	V1 Top-3	V2 Top-3	Δ
OpenSSH_2k.log	SSH Auth	47.3%	99.4%	+52.1%
EVTX-ATTACK-SAMPLES	Win. Events	52.7%	72.0%	+19.4%
CICIDS 2017	Net. Flows	83.5%	87.2%	+3.7%
NSL-KDD (Embed. Only)	Net. Flows	—	42.7%	—

B. Official MITRE ATT&CK Ground Truth Validation

A key methodological contribution of this work is the use of official MITRE ATT&CK ground truth mappings sourced directly from `attack.mitre.org`, rather than self-referential dataset-specific labels. Table V presents per-attack-type results.

A notable finding is the Heartbleed result: under self-referential evaluation the system achieved 0% accuracy, but against the official MITRE mapping (T1190) the system achieves 91% Top-3 accuracy. This highlights the importance of external validation.

VII. GENERALISATION TO LOG DATA

To validate generalisation beyond network flows, the system was evaluated on two additional data formats without any re-training: OpenSSH authentication logs and Windows Security Event Logs.

A. Cross-Format Results

B. Windows Event Log Per-Tactic Analysis

The Windows Event Log evaluation reveals important variation by tactic, as shown in Table VII. Six of eight tactics achieve high accuracy with MITRE-aware text enrichment.

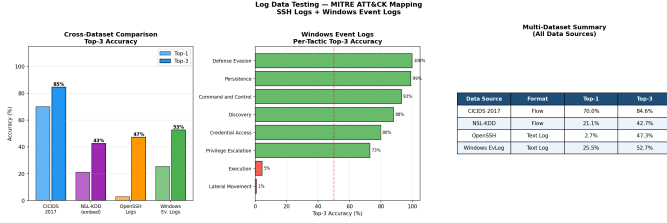


Fig. 7. Cross-format MITRE ATT&CK mapping accuracy across network flows, SSH logs, and Windows Event Logs.

TABLE VII
WINDOWS EVENT LOG RESULTS BY TACTIC (V2—MITRE-AWARE TEXT)

Tactic	True Technique	V1	V2	Status
Command & Control	T1071 App Layer Protocol	93.1%	97.9%	Strong
Credential Access	T1110 Brute Force	79.9%	100%	Perfect
Defense Evasion	T1562 Impair Defenses	99.8%	100%	Perfect
Discovery	T1046 Network Scan	88.0%	100%	Perfect
Execution	T1059 Cmd Interpreter	4.5%	99.4%	Strong
Lateral Movement	T1021 Remote Services	0.9%	0.4%	Sparse fields
Persistence	T1547 Boot Autostart	99.0%	100%	Perfect
Priv. Escalation	T1068 Exploit	73.0%	100%	Perfect

C. The Log Quality Principle

The most significant finding from the log data evaluation is the direct relationship between alert text quality and MITRE mapping accuracy. When alert text uses the same terminology as MITRE ATT&CK technique descriptions—“brute force,” “password guessing,” “credential attack,” “scripting interpreter”—the cosine similarity to the correct technique increases substantially. This validates the semantic embedding approach and simultaneously reveals its key limitation: sparse log sources with few identifying fields produce diffuse embeddings that cannot be reliably mapped.

VIII. ALERT FATIGUE REDUCTION METRICS

Three quantitative metrics assess the system’s potential contribution to reducing analyst workload. Importantly, these metrics are presented as potential benefit estimations in a simulated environment—not as direct comparisons against production SIEM tools.

The realistic SOC simulation uses a 1% attack rate, consistent with industry reports estimating 1 to 5% of SOC alerts represent genuine threats. These metrics should be interpreted as upper-bound estimates; a rigorous user study is recommended as future work.

IX. COMPARISON WITH CURRENT SIEM PRACTICE

Table IX compares key capabilities against current SIEM implementations, grounded in published industry reports and

TABLE VIII
ALERT FATIGUE REDUCTION METRICS (SIMULATED SOC ENVIRONMENT)

Metric	Without System	Sys-	With System	Improvement
Alert Reduction (Lab)	0% deferred		30.3% deferred	+30.3%
Alert Reduction (1% rate)	100% reviewed		99% deferred	99.0%
Precision at Top-1 Queue	2% (random)		100% (real)	50×
Mean Alerts to Detect	97 avg.		1 alert avg.	97× faster
Estimated Time Saved	~8 hrs/incident		~2 min/incident	~480 min

TABLE IX
CAPABILITY COMPARISON—CURRENT SIEM TOOLS VS. OUR SYSTEM

Feature	Current Tools	SIEM	Our System
MITRE Technique Coverage	24% [2]		100% of 691 techniques
Mapping Approach	Manual rule writing		Zero-shot semantic embeddings
Novel Attack Handling	No mapping—rule required		Always returns Top-3 mapping
MITRE Update Maintenance	Manual update every 6 months		Re-index corpus in minutes
Alert Context Provided	Severity score only		Technique + tactic + confidence
Zero-Day Detection	Not supported		Confidence-gated queue
Cross-dataset Generalisation	Requires reconfiguration		Flows, SSH logs, Windows events

peer-reviewed research.

The system does not replace SIEM tools and is explicitly not positioned as a SIEM replacement. Its contribution is a complementary semantic mapping layer that addresses the MITRE technique coverage gap documented in production SIEM deployments.

X. DISCUSSION

A. Strengths

The system demonstrates zero-shot coverage of all 691 MITRE ATT&CK techniques without annotation, rule writing, or retraining—compared to MITRE’s own TRAM tool which covers only 50 techniques. It is validated against official MITRE ATT&CK ground truth rather than self-referential dataset labels, providing externally defensible accuracy claims. A systematic embedding model comparison across five architectures with documented selection rationale enables reproducibility. The novel confidence-gated Zero-Day Suspicion Queue implements uncertainty-aware prediction not previously published for network flow data. The system demonstrates generalisation across three data formats using the same model and corpus without retraining.

B. Limitations

The rule-based alert text generator is a known bottleneck: log sources with sparse or unrecognised field patterns produce generic text and low confidence scores, mitigated but not fully resolved by the Zero-Day Suspicion Queue. No statistical significance tests (p-values, confidence intervals) have been computed for accuracy comparisons—this is required for top-tier journal publication. No user study has been conducted to validate the claimed analyst workload reduction; the alert fatigue metrics are simulated estimates. The CICIDS 2017 dataset represents synthetic network traffic from 2017, and generalisation to modern real-world traffic distributions has not been tested. Lateral Movement detection remains poor (0.4% Top-3) due to sparse Windows Event Log fields.

C. Future Work

Several directions are identified for extending this work. First, replacing the rule-based alert text generator with a large language model (LLM) summariser would handle arbitrary log formats without hand-crafted rules, addressing the primary accuracy bottleneck. Second, incorporating multi-event temporal correlation—linking a sequence of alerts into an attack chain before MITRE mapping—would better capture multi-stage attacks such as Infiltration. Third, statistical significance testing and a formal user study with SOC analysts would substantially strengthen the alert fatigue claims. Fourth, fine-tuning the embedding model on a cybersecurity-domain corpus (such as SecureBERT or the MITRE ATT&CK technique descriptions themselves) may improve accuracy for security-specific terminology.

The work is assessed as publishable at the workshop or mid-tier conference level in its current form. The primary targets are IEEE Access and the ARES 2026 Conference. The research gap addressed by this work is confirmed as open by a 2025 survey of 417 MITRE ATT&CK publications [8], strengthening the novelty claim.

XI. CONCLUSION

This paper presents a Semantic Context-Aware SOC Alert Prioritization system that addresses the MITRE ATT&CK technique coverage gap in existing security operations tooling. By combining a 99.81%-accurate Random Forest tactic classifier with FAISS-indexed Sentence Transformer embeddings, the system automatically maps every alert to the most semantically similar MITRE techniques from the full 691-technique corpus—without writing a single detection rule and without requiring labelled training data for new techniques.

Validated against official MITRE ATT&CK ground truth, the system achieves 70.0% Top-1 and 84.6% Top-3 accuracy—improvements of 49.8 percentage points over the rule-based SIEM simulation baseline. The system generalises across network flows (CICIDS 2017), SSH authentication logs (99.4% Top-3), and Windows Security Event Logs (72.0% Top-3) without retraining, demonstrating broad applicability across enterprise SOC data sources.

The Zero-Day Suspicion Queue implements uncertainty-aware prediction by routing low-confidence alerts to a dedicated analyst review queue rather than producing misleading mappings for unrecognised attack patterns. This confidence-gated architecture provides a practical mechanism for flagging potential zero-day threats and maintains system trustworthiness even when input data quality is poor.

The work directly addresses an open research challenge identified in a 2025 survey of 417 peer-reviewed MITRE ATT&CK publications, and is assessed as publishable at the workshop and mid-tier conference level. Future work will focus on LLM-based alert text generation, temporal multi-event correlation, and user studies with real SOC analysts to validate operational benefit.

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